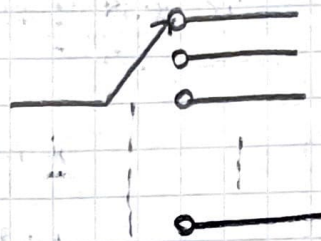
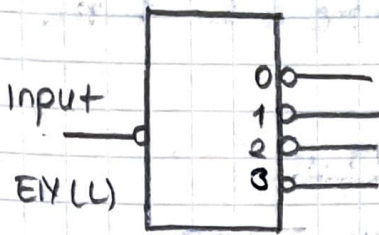
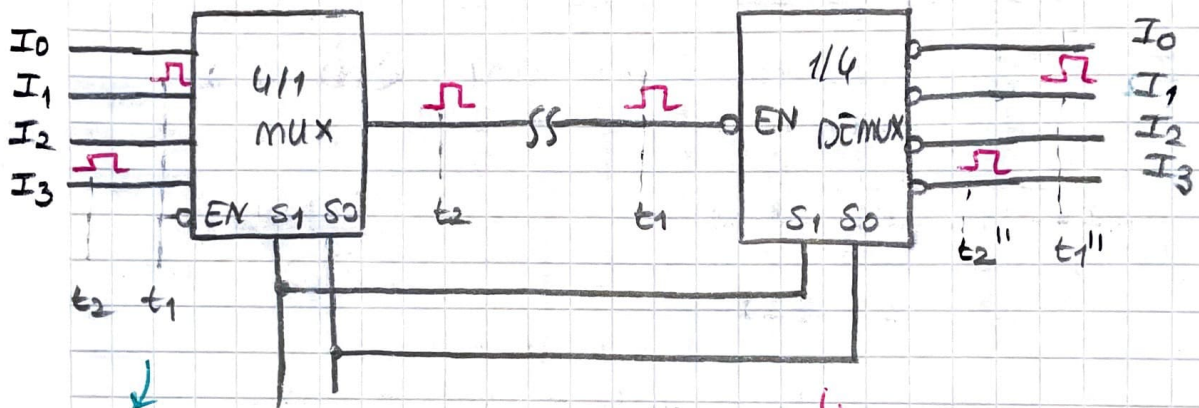


DEMULTIPLEXER



select switch



↓
Tek bir hatt
üzerinden paylaşımlı
olarak sinyal gönderimi.

↳ Paralel hat kullanılmakla
bu şekilde yapıldığında
maliyet düşer.

TYPES OF CODES

NUMERIC CODES

- Weighted Binary Codes
- Binary Coded Decimal (BCD)
- Self Complementing Codes
- Reflective Codes
- Unit Distance Codes
- Unit Distance Reflective Codes
- Error Detecting Codes
- Error Detecting and Correcting Codes

ALPHANUMERIC CODES

- EBCDIC
- ASCII

BCD $\rightarrow$ 10'luk sistemdeki her bir hane 4 bit olarak ifade edilir.

<u>Decimal</u>	<u>Binary</u>	<u>NBCD</u>
10	1010	<u>0001</u> <u>0000</u> 1 0
141	10001101	<u>0001</u> <u>0100</u> <u>0001</u> 1 4 1
375	101110111	<u>0011</u> <u>0111</u> <u>0101</u> 3 7 5

WEIGHTED BCD CODES

$$8421 = \text{NBCD}$$

$$\rightarrow N = W_3 \cdot X_3 + W_2 \cdot X_2 + W_1 \cdot X_1 + W_0 \cdot X_0$$

W : Weighting factor

Excess $\leftarrow$ XS3 $\rightarrow$ 3 fazlalık kodu

2421 $\rightarrow$ Aiken

5421

5311

7421

9437

8421 $\rightarrow$ negatif etki

86421

51111

$\downarrow$

0'dan değil 3'ten başlar
yani mesela

0, 0 ile değil 3 ile

1, 1 ile değil 4 ile

2, 2 ile değil 5 ile gösterilir.

Dec	Natural BCD	+3 Kodu	Aiken	Bu sayılar ağırlık
	NBCD	XS3	2421	51111
0	0000	0011	0000	00000
1	0001	0100	0001	00001
2	0010	0101	0010	00011
3	0011	0110	0011	00111
4	0100	0111	0100	01111
5	0101	1000	0101	10000
6	0110	1001	0110	11000
7	0111	1010	0111	11100
8	1000	1011	1110	11110
9	1001	1100	1111	11111

↓
Sıralabilir
↓
Self complementing coding

→ Neden +3 kodu var? Çünkü 0000 girişi yok, hata tespiti daha kolay.

SELF COMPLEMENTING BCD CODES

→ XS3
6317
2421 (Aiken)

Dec	XS3	2421	Reflected Code			
			A	B	C	D
0	0011	0000	0	0	0	0
1	0100	0001	0	0	0	1
2	0101	0010	0	0	1	0
3	0110	0011	0	0	1	1
4	0111	0100	0	1	0	0
5	1000	0101	1	1	0	0
6	1001	0110	1	0	1	1
7	1010	0111	1	0	1	0
8	1011	1110	1	0	0	1
9	1100	1111	1	0	0	0

Reflected Code → Birbirini 9'a tamamlayan kod. Sözcükler arasındaki uzaklık 1 old. kodlardır.

Logikte uzaklık → Kod sözcükleri arasındaki hane değişikliği sayısıdır.

1011
1100 } 3

0000
0001 } 1

uzaklık
1

0000
1001 } 2

uzaklık
2

?

UNIT DISTANCE CODES (UDC)

Dec	UD BCD	4 BIT GRAY	3BIT GRAY
0	0000	0000	000
1	0001	0001	001
2	0011	0011	011
3	0010	0010	010
4	0110	0110	110
5	1110	0111	111
6	1111	0101	101
7	1101	0100	100
8	1100	1100	
9	0100	1101	
10		1111	
11		1110	
12		1010	
13		1011	
14		1001	
15		1000	

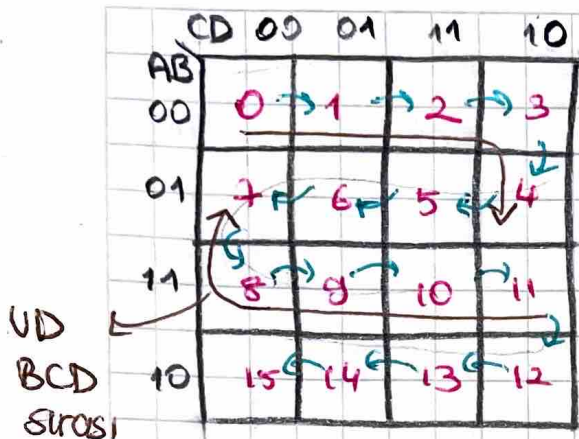
Unit code → Ard arda tüm kod sözcüklerinin arasındaki uzaklık 1 olması gerekir.

BC	00	01	11	10
A	0 0	1 1	2 3	3 2
1	7 4	6 5	5 7	4 6

3 BIT GRAY CODE

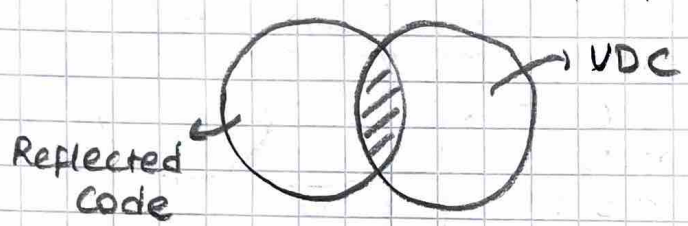
- 000 → 0
- 001 → 1
- 011 → 2
- 010 → 3
- 110 → 4
- 111 → 5
- 101 → 6
- 100 → 7

→ Gray kodu kolay hatırlama,

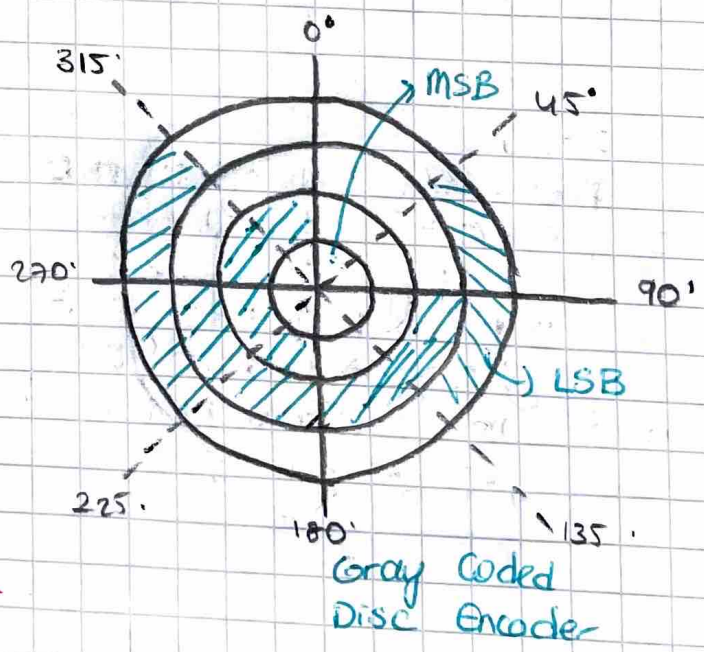
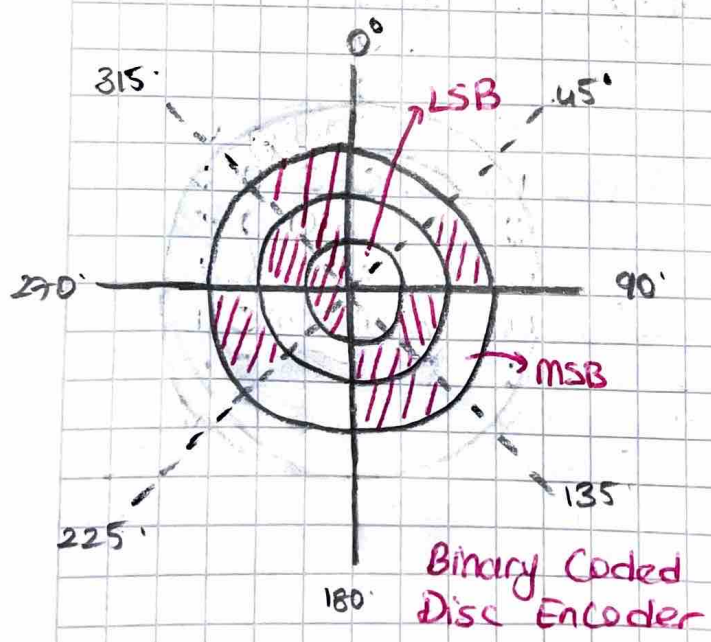


4 BIT CODE

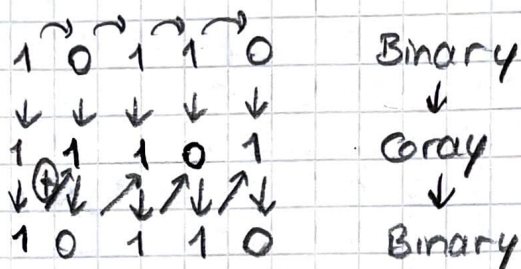
4 → 0110
 5 → 0111



ANGLE	Binary Code	GRAY Code
0°	000	000
45°	001	001
90°	010	011
135°	011	010
180°	100	110
225°	101	111
270°	110	101
315°	111	100



Binary to Gray Code Conversion Gray to Binary Code Conversion



ERROR DETECTING CODES

Parity $\left\{ \begin{array}{l} \text{Even Parity} \\ \text{Odd Parity} \end{array} \right.$

Word Odd Parity Even Parity

1001 11001 01001
 1110 01110 11110
 1101 11101 01101 yanlış!

↓ ↓
 0 olmalı 1 olmalı

Dec	NBCD	Even Parity	Odd Parity	5 0	4 3 2 1 0
0	0000	00000	10000	01	00001
1	0001	10001	00001	01	00010
2	0010	10010	00010	01	00100
3	0011	00011	10011	01	01000
4	0100	10100	00100	01	10000
5	0101	00101	10101	10	00001
6	0110	00110	10110	10	00010
7	0111	10111	00111	10	00100
8	1000	11000	01000	10	01000
9	1001	01001	11001	10	10000

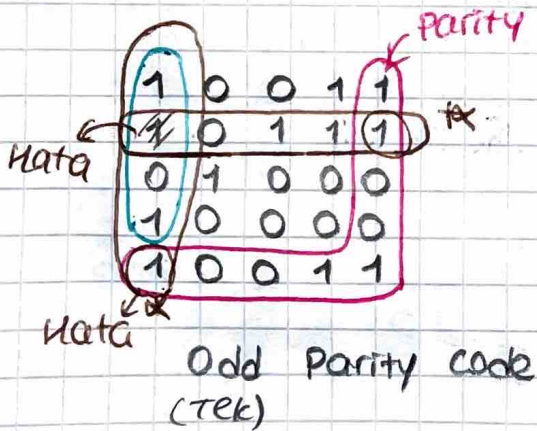
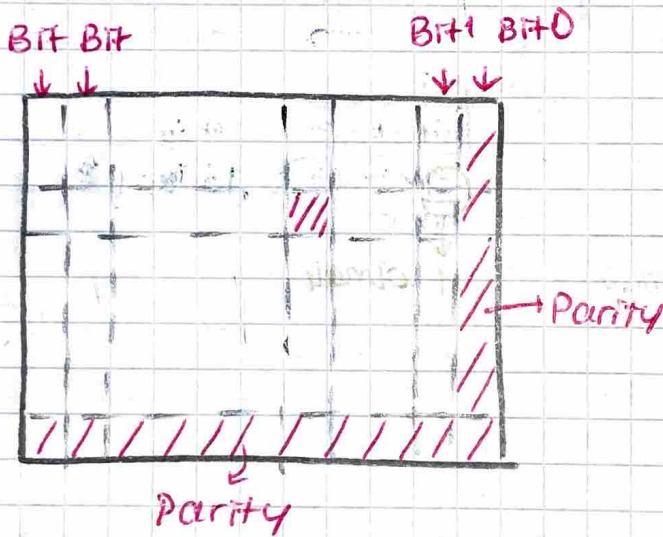
↓ ↓
 Çift Parity Tek Parity

Odd Parity → Parity biti dahil 1'lerin sayısı tek olacak şekilde

Even Parity → Parity biti dahil 1'lerin sayısı çift olacak şekilde

Dec	One-Hot Ring Counter									
	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	1
1	0	0	0	0	0	0	0	0	1	0
2	0	0	0	0	0	0	0	1	0	0
3	0	0	0	0	0	0	1	0	0	0
4	0	0	0	0	0	1	0	0	0	0
5	0	0	0	0	1	0	0	0	0	0
6	0	0	0	1	0	0	0	0	0	0
7	0	0	1	0	0	0	0	0	0	0
8	0	1	0	0	0	0	0	0	0	0
9	1	0	0	0	0	0	0	0	0	0

Error Detecting and Correcting Code



→ Odd Parity'le göre bir yatay bir dikey bakıyoruz.

→ İki satırın kesişimindeki hücrede hata oldu. eviriyoruz. (1 bitlik hata)

→ Aynı anda 2 bitlik hata olmuş ise bu kod tespit edemez.

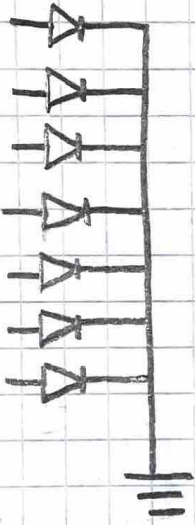
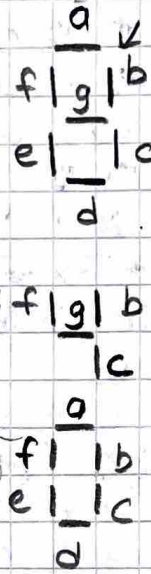
→ Hocam: bu konuda
ciddiyetle örnekler x

Seven Segment Code

Dec | a b c d e f g

0	1	1	1	1	1	0
1	0	1	1	0	0	0
2	1	1	0	1	1	0
3	1	1	1	1	0	0
4	0	1	1	0	0	1
5	1	0	1	1	0	1
6	0	0	1	1	1	1
7	1	1	1	0	0	0
8	1	1	1	1	1	1
9	1	1	1	0	0	1

Comon Cathode
Comon Anode



Common cathode

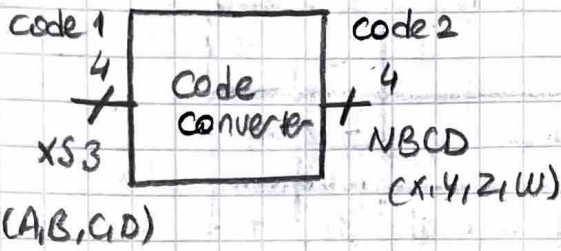
Common Cathode şeklinde
Common Anode bunun tam
tersi

→ Common Anode da var. Onda uçlar birleştirilir.

→ Sınarda standardi bozma.

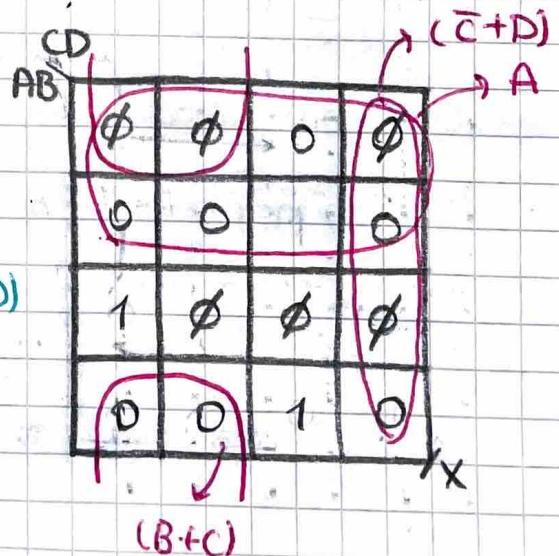
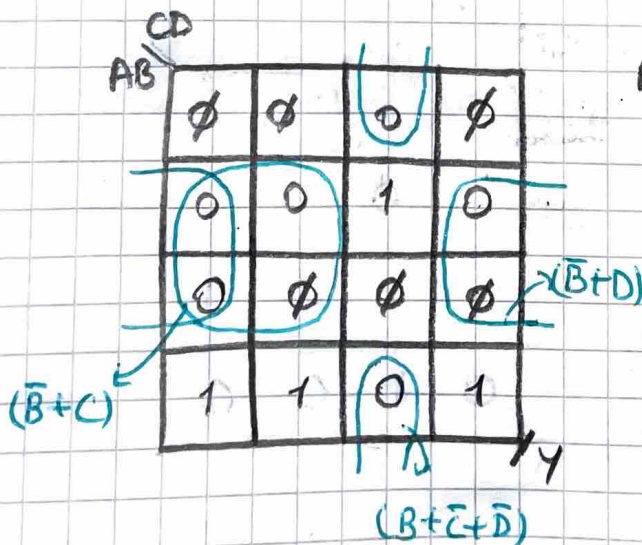
ÖRN

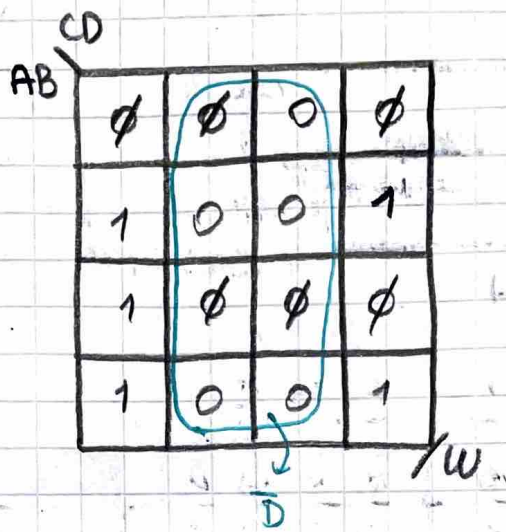
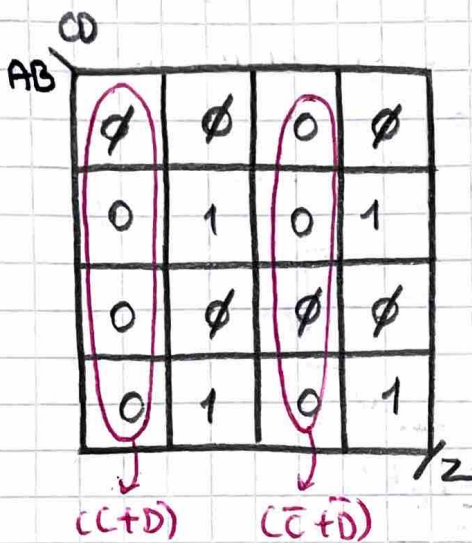
İki fazlalıklı kodlara (XS3), NBCD (Doğal BCD) kodu elde eden bir devre tasarlayınız.



- Sadece NOR kapılarını kullan.
- Min. maliyetle gerçekleştir.

Dec	A	B	C	D	X	Y	Z	W
0	0	0	0	0				
1	0	0	0	1				
2	0	0	1	0				
3	0	0	1	1	0	0	0	0
4	1	0	1	0	0	0	0	1
5	2	0	1	0	0	0	1	0
6	3	0	1	1	0	0	1	1
7	4	0	1	1	0	1	0	0
8	5	1	0	0	0	1	0	1
9	6	1	0	0	0	1	1	0
10	7	1	0	1	0	1	1	1
11	8	1	0	1	1	0	0	0
12	9	1	1	0	1	0	0	1
13	1	1	0	1				
14	1	1	1	0				
15	1	1	1	1				



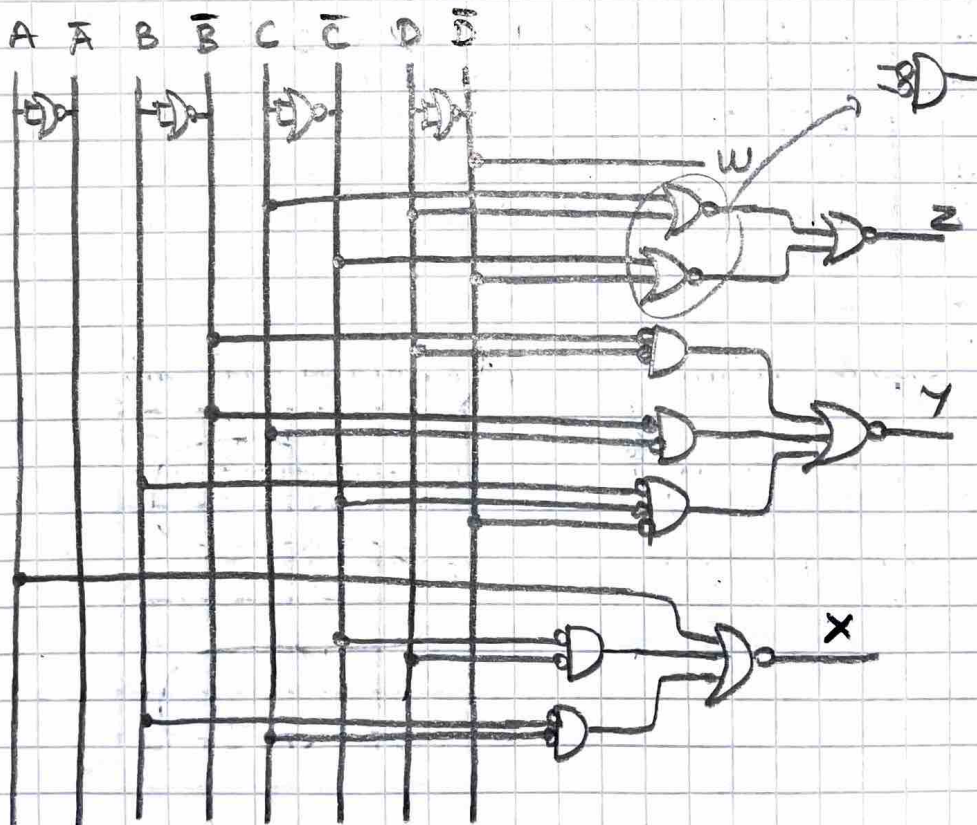


$$X = A \cdot (\bar{C}+D) \cdot (B+C)$$

$$Y = (\bar{B}+D) \cdot (\bar{B}+C) \cdot (B+\bar{C}+\bar{D})$$

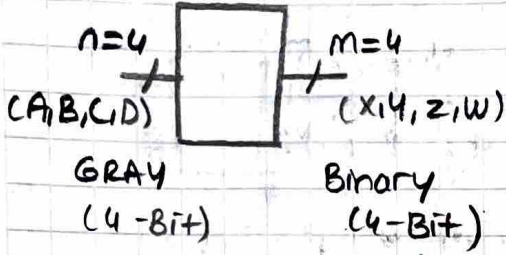
$$Z = (C+D) \cdot (\bar{C}+\bar{D}) \rightarrow CC$$

$$W = \bar{D}$$

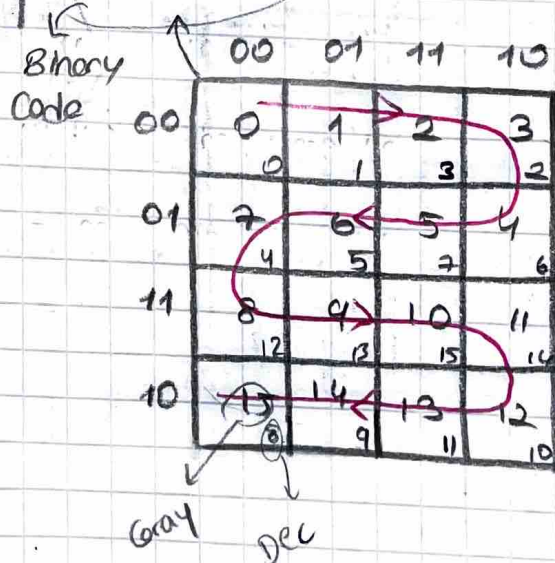


ÖRNEK

4 bitlik GRAY kodun ikili koda dönüşümünü sağlayan bir kod dönüştürücüyü (4/16) erimen çıkışlı bir decoder kullanarak tasarlayın. Çıkışın aktiflik düzeyi logic 1 olacaktır.



Dec	A	B	C	D	X	Y	Z	W
0	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0	1
2	0	0	1	0	0	0	1	1
3	0	0	1	1	0	0	1	0
4	0	1	0	0	0	1	1	1
5	0	1	0	1	0	1	1	0
6	0	1	1	0	0	1	0	0
7	0	1	1	1	0	1	0	1
8	1	0	0	0	1	1	1	1
9	1	0	0	1	1	1	1	0
10	1	0	1	0	1	1	0	0
11	1	0	1	1	1	1	0	1
12	1	1	0	0	1	0	0	0
13	1	1	0	1	1	0	0	1
14	1	1	1	0	1	0	1	1
15	1	1	1	1	1	0	1	0



$$X = m_8 + m_9 + m_{10} + m_{11} + m_{12} + m_{13} + m_{14} + m_{15}$$

$$\overline{X} = \overline{m_8 \cdot m_9 \cdot m_{10} \cdot m_{11} \cdot m_{12} \cdot m_{13} \cdot m_{14} \cdot m_{15}}$$

$$Y = m_4 + m_5 + m_6 + m_7 + m_8 + m_9 + m_{10} + m_{11}$$

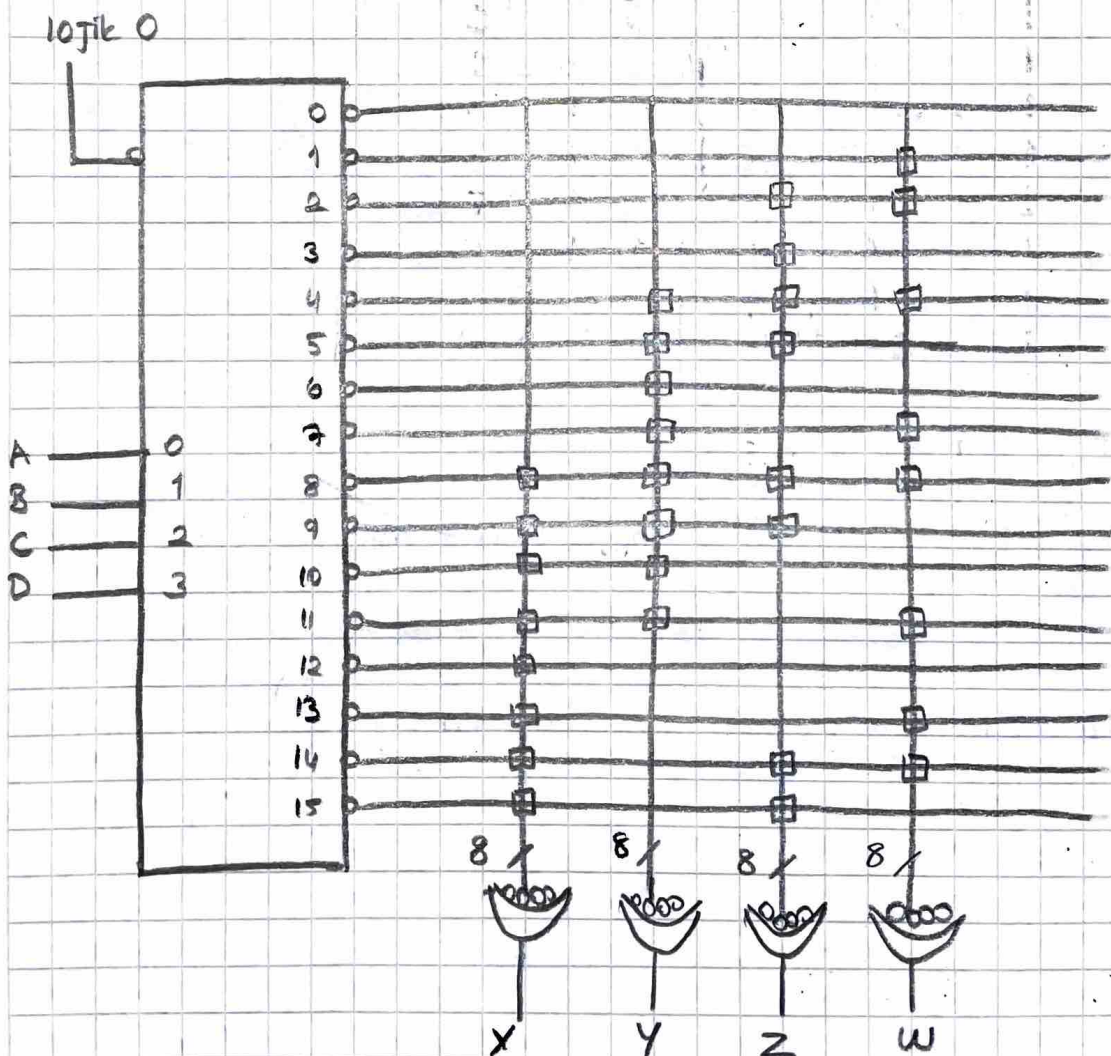
$$\overline{Y} = \overline{m_4 \cdot m_5 \cdot m_6 \cdot m_7 \cdot m_8 \cdot m_9 \cdot m_{10} \cdot m_{11}}$$

$$Z = m_2 + m_3 + m_4 + m_5 + m_8 + m_9 + m_{14} + m_{15}$$

$$\overline{Z} = \overline{m_2 \cdot m_3 \cdot m_4 \cdot m_5 \cdot m_8 \cdot m_9 \cdot m_{14} \cdot m_{15}}$$

$$W = m_1 + m_2 + m_4 + m_7 + m_8 + m_{11} + m_{13} + m_{14}$$

$$\overline{W} = \overline{m_1 \cdot m_2 \cdot m_4 \cdot m_7 \cdot m_8 \cdot m_{11} \cdot m_{13} \cdot m_{14}}$$



MULTIPLIER (Garp101)



Dec	B	A	B ₁	B ₀	A ₁	A ₀	B x A	P ₃	P ₂	P ₁	P ₀
0	0	0	0	0	0	0	0	0	0	0	0
1	0	1	0	0	0	1	0	0	0	0	0
2	0	2	0	0	1	0	0	0	0	0	0
3	0	3	0	0	1	1	0	0	0	0	0
4	1	0	0	1	0	0	0	0	0	0	0
5	1	1	0	1	0	1	1	0	0	0	1
6	1	2	0	1	1	0	2	0	0	1	0
7	1	3	0	1	1	1	3	0	0	1	1
8	2	0	1	0	0	0	0	0	0	0	0
9	2	1	1	0	0	1	2	0	0	1	0
10	2	2	1	0	1	0	4	0	1	0	0
11	2	3	1	0	1	1	6	0	1	1	0
12	3	0	1	1	0	0	0	0	0	0	0
13	3	1	1	1	0	1	3	0	0	1	1
14	3	2	1	1	1	0	6	0	1	1	0
15	3	3	1	1	1	1	9	1	0	0	1

$$P_3 = \sum m(15) = m_{15}$$

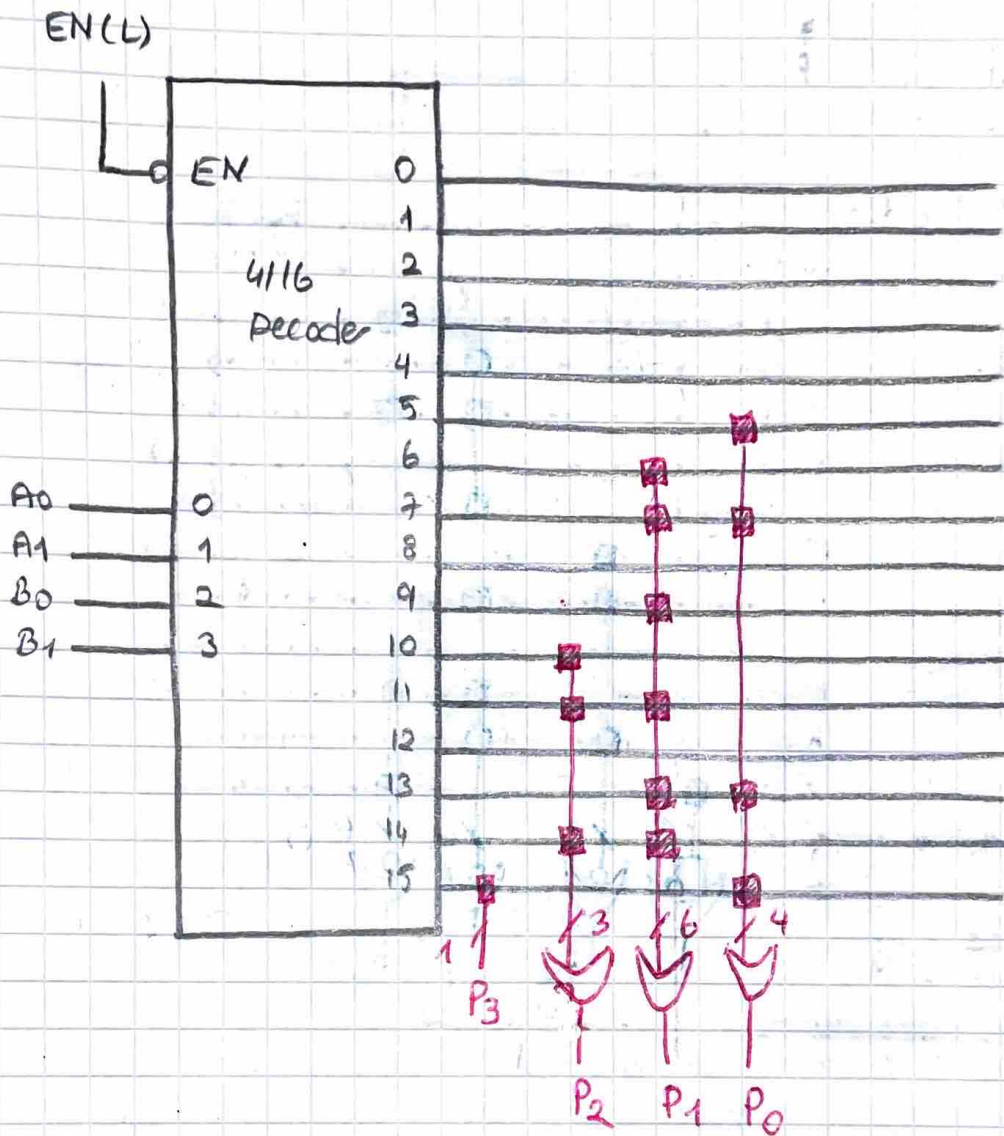
$$P_2 = \sum m(10, 11, 14) = m_{10} + m_{11} + m_{14}$$

$$P_1 = \sum m(6, 7, 9, 11, 13, 14) = m_6 + m_7 + m_9 + m_{11} + m_{13} + m_{14}$$

$$P_0 = \sum m(5, 7, 13, 15) = m_5 + m_7 + m_{13} + m_{15}$$

→ minterm

a) Evirmeyen çıkışlı 4/16 Decoder ile gerçekleştirin.



$$P_3 = m_{15}$$

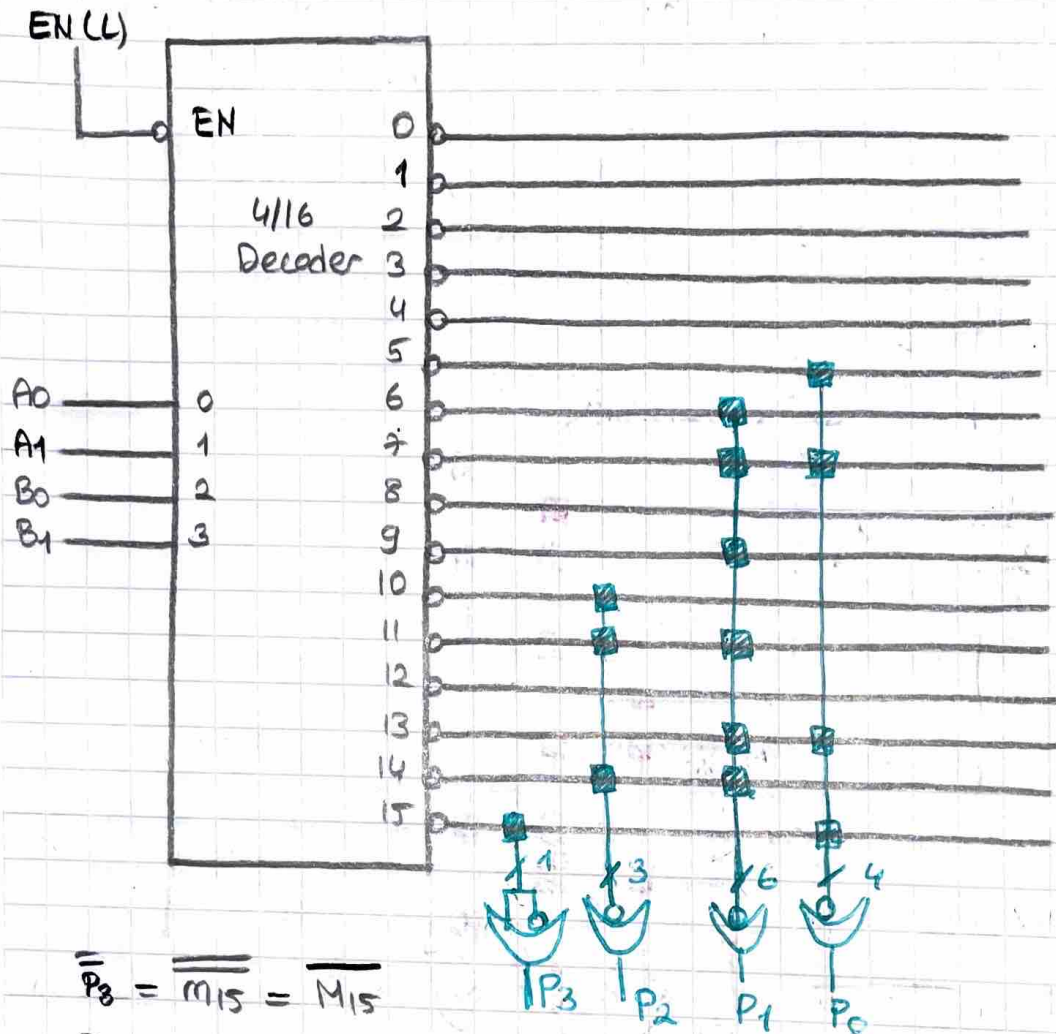
$$P_2 = m_{10} + m_{11} + m_{14}$$

$$P_1 = m_6 + m_7 + m_{11} + m_{13} + m_{14}$$

$$P_0 = m_5 + m_7 + m_{13} + m_{15}$$

maxterm

b) Eviren çıkışlı 4/16 Decoder ile



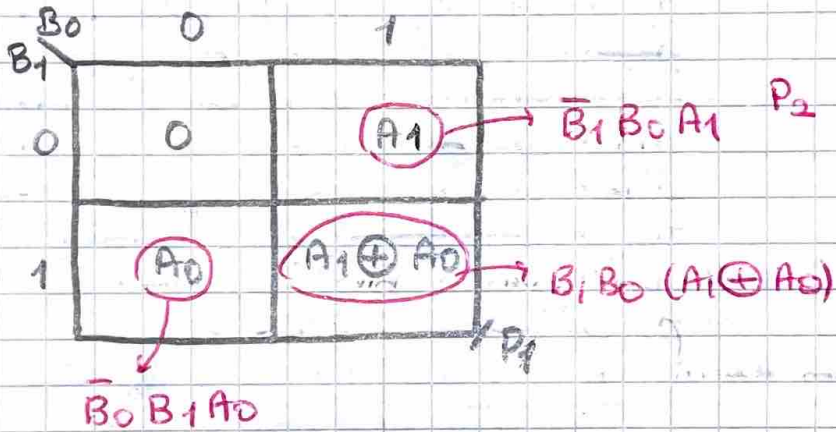
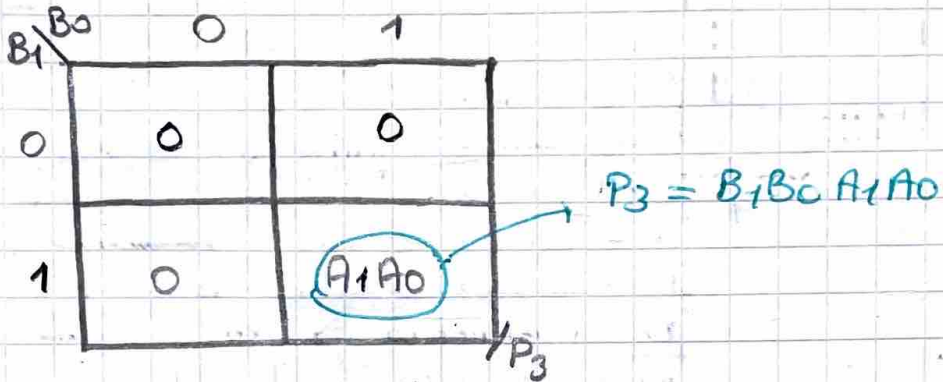
$$\overline{P_3} = \overline{m_{15}} = \overline{M_{15}}$$

$$\overline{P_2} = \overline{m_{10} + m_{11} + m_{14}} = \overline{M_{10} \cdot M_{11} \cdot M_{14}}$$

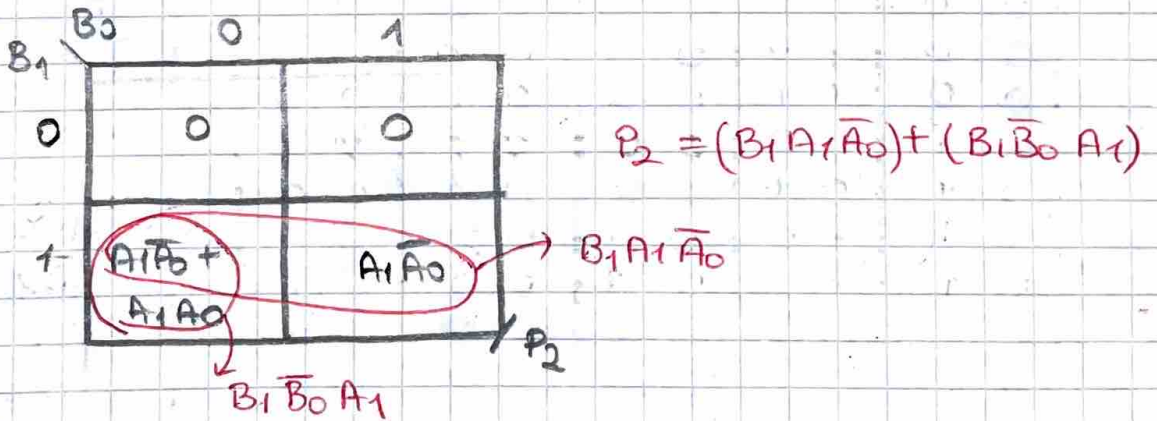
$$\overline{P_1} = \overline{m_6 + m_7 + m_9 + m_{11} + m_{13} + m_{14}} = \overline{M_6 \cdot M_7 \cdot M_9 \cdot M_{11} \cdot M_{13} \cdot M_{14}}$$

$$\overline{P_0} = \overline{m_5 + m_7 + m_{13} + m_{15}} = \overline{M_5 \cdot M_7 \cdot M_{13} \cdot M_{15}}$$

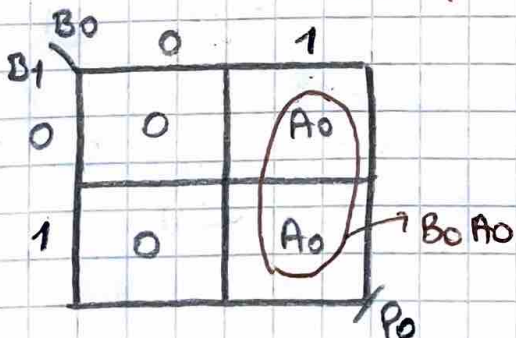
2. mertebeden K-map ile



$$P_2 = (B_1 \bar{B}_0 A_0) + (\bar{B}_1 B_0 A_1) + (B_1 B_0 (A_1 \oplus A_0))$$

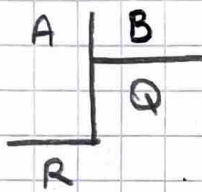
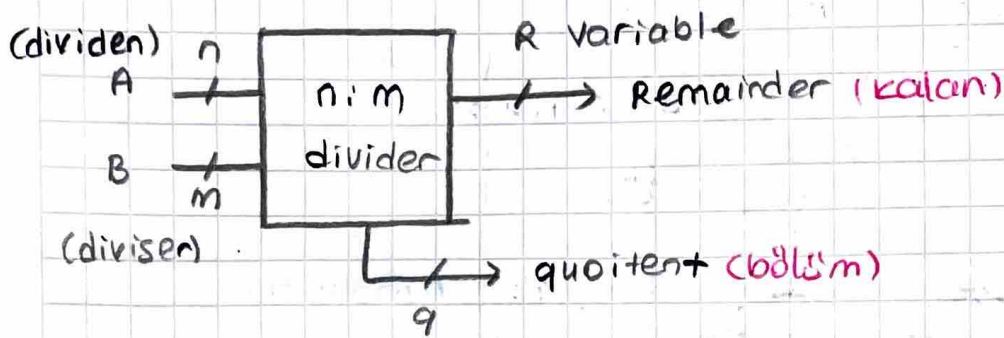


$$P_2 = (B_1 A_1 \bar{A}_0) + (B_1 \bar{B}_0 A_1)$$



$$P_0 = B_0 A_0$$

DIVIDER (bölücü)



$$\frac{0}{0} \text{) Belirsiz } \Rightarrow \emptyset$$

$$\frac{A}{0} \text{) Sonsuz } \Rightarrow \emptyset$$

$$\frac{0}{B} \Rightarrow 0$$

$$A = B \cdot Q + R$$

$\nearrow$ bölüm $\nearrow$ kalan

Binary

Dec	A ₁	A ₀	B ₁	B ₀	Q	R	Q ₁	Q ₀	R ₁	R ₀	Q ₁	Q ₀	R ₁	R ₀
0	0	0	0	0	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	0	0	0	0
1	0	0	0	1	0	0	0	0	0	0				
2	0	0	1	0	0	0	0	0	0	0				
3	0	0	1	1	0	0	0	0	0	0				
4	0	1	0	0	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	—	—	—	—
5	0	1	0	1	1	0	0	1	0	0				
6	0	1	1	0	0	1	0	0	0	1	0	$\bar{B}_1 \cdot B_0$	0	$B_1 \cdot \bar{B}_0 + B_1 \cdot B_0$
7	0	1	1	1	0	1	0	0	0	1			(B ₁)	(B ₁)
8	1	0	0	0	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	—	—	—	—
9	1	0	0	1	2	0	1	0	0	0	$\bar{B}_1 \cdot B_0$	$B_1 \cdot \bar{B}_0$	$B_1 \cdot B_0$	0
10	1	0	1	0	1	0	0	1	0	0	(B ₁)	(B ₀)		
11	1	0	1	1	0	2	0	0	1	0				
12	1	1	0	0	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	—	—	—	—
13	1	1	0	1	3	0	1	1	0	0	$\bar{B}_1 \cdot B_0$	1	0	$B_1 \cdot \bar{B}_0$
14	1	1	1	0	1	1	0	1	0	1	(B ₁)			
15	1	1	1	1	1	0	0	1	0	0				

2. mertebeden K- Map ile

$$\begin{aligned} N &= 4 \\ n &= 2 \end{aligned} \quad \left. \begin{array}{l} N-n = 2 \quad (B_1, B_0) \\ 2^{N-n} = 2^2 = 4 \end{array} \right\}$$

	A ₀	0	1
A ₁			
0	0		0
1	$\bar{B}_1 B_0$	$\bar{B}_1 \bar{B}_0$	

$Q_1 = A_1 \bar{B}_1$

	A ₀	0	1
A ₁			
0	0		$\bar{B}_1$
1	$\bar{B}_0$	$B_1 \bar{B}_1$	$B_0 \bar{B}_0$

$Q_0 = A_1 \bar{B}_0 + A_0 \bar{B}_1 + A_1 A_0$

	A ₀	0	1
A ₁			
0	0		0
1	$B_1 B_0$		0

$R_1 = A_1 \bar{A}_0 B_1 B_0$

	A ₀	0	1
A ₁			
0	0		$B_1 \bar{B}_0 + B_1 B_0$
1	0		$B_1 \bar{B}_0$

$R_0 = \bar{A}_1 A_0 B_1 + A_1 A_0 \bar{B}_0$

	A ₀	0	1
A ₁			
0	0		B_1
1	0		$\bar{B}_0$

$A_1 A_0 \bar{B}_0$

4x1 MUX ve yalnızca NAND kapıları kullanarak
min maliyetle gerçekleştirin.

$A_1 \bar{A}_1 A_0 \bar{A}_0 \quad B_1 \bar{B}_1 B_0 \bar{B}_0$

